



Amazing Flying Warehouse For Delivery From The Sky

This article covers some technologies being considered for the logistics involved in quick delivery of the goods through future drone delivery systems, including a flying warehouse up in the sky

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The advancements in new technologies have changed the world to a large extent. In recent years, the information and communications technology (ICT), artificial intelligence, drones, and robotics have become the focus of intense development, which has led to active convergence between seemingly unrelated sectors.

Logistics is one such sector that is enjoying the recent advancements in technologies. The Internet of Things (IoT) and ICT have automated logistics systems to ensure that all processes from customer order to delivery are performed with hundred percent precision. This allows customers to check delivery status in real time and receive the ordered goods within a few hours.

Another revolution is now underway courtesy drones. The drone delivery system ensures economical and swift services because of maneuverability, flexibility, and low cost. Several logistics companies like DHL, Amazon, FedEx, and Alibaba have introduced drones for goods delivery to their customers. Amazon launched its unmanned aerial vehicle (UAV) delivery

service in the UK in December 2016 to deliver packages within thirty minutes of receipt of an order.

A UAV delivery service, however, has some technological limitations. Drones can fly for a limited time only due to power constraints of their lithium batteries, which limits their scalability and serviceability. The logistics companies are therefore developing new systems to deal with the limitation so that they can provide faster service to more customers. One state-of-the-art system is the airborne fulfilment centre (AFC) delivery system of Amazon.

The AFC delivery system was first revealed in April 2016 when Amazon patented a flying warehouse that would deploy UAVs to deliver parcels to customers. In this system, an AFC would store goods as well as small UAVs in the sky at an altitude close to the stratosphere.

When an order is received, a UAV carrying the ordered items would be deployed from the AFC to the customer, as shown in Fig. 1(a). It will return to an Amazon material handling facility (MHF) after performing

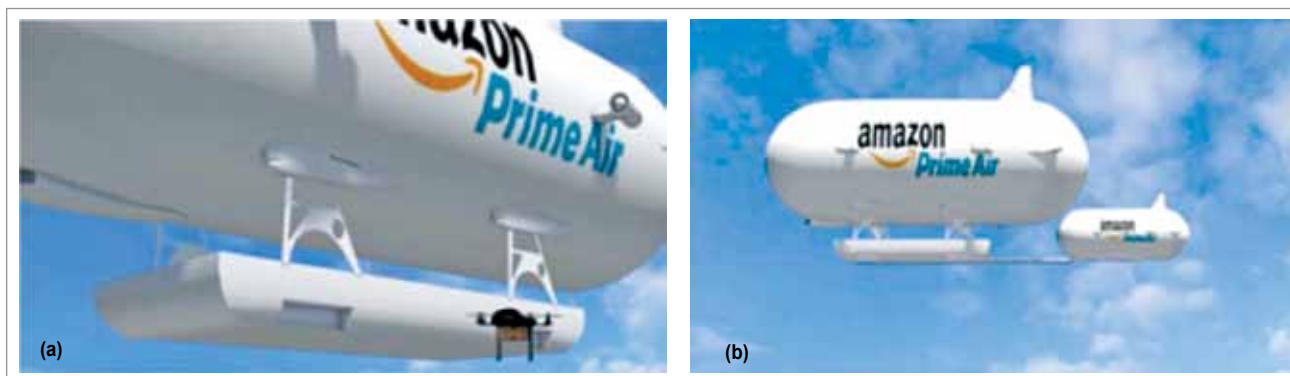


Fig. 1(a) and (b): Amazon AFC delivery system